Department of Computer and Information Systems Engineering

CS-406 Computer Engineering Project

Proposal for the Final Year Design Project

Title	MeshX : Adaptive Off-Grid Communication System using LoRa

Domain	**Domain 1** Embedded Systems	**Domain 2** Networking	**Domain 3** Software Engineering	**Domain 4** Mobile App Development	**Domain 5**	**Domain 6**

1. Nature of Project [Tick all that applicable]

☒ New Project OR ☐ Extension of Existing Project	☒ Industrial Collaboration	☐ Funded
☐ Other Department Collaboration (If yes) Department Name_____	☐ Other Academic Institution Collaboration (If yes) Institution Name_____	

2. Brief Outline (*Problem Identification and Significance*)

During disasters, large public gatherings, or in remote/rural terrain, cellular networks and Wi-Fi infrastructure are frequently unavailable, congested, or destroyed, cutting off affected communities from emergency communication exactly when it matters most. Existing solutions either depend on satellite hardware that is too expensive for civilian/student deployment, or on single-hop radios with severely limited range.

This project proposes a low-cost, battery-powered LoRa mesh network of 5-6 nodes that lets people exchange text messages over several kilometers without any cellular network, Wi-Fi, or internet connection. Each node forwards messages for its neighbors, so two people far apart can still reach each other through several intermediate hops and, unlike a simple flooding-based mesh, this system will make routing decisions using measured link quality (signal strength, noise, packet loss, and node battery level) rather than blind hop-count, which is the project's core original contribution.

3. Objectives

The major objectives to be accomplished by this project are:

- To design and build a 5-6 node LoRa mesh hardware platform (ESP32-S3 + SX1262 transceiver) with battery power, charging protection, and a phone-facing BLE interface.
- To implement a custom mesh networking protocol from scratch, node discovery, multi-hop forwarding, TTL/hop limiting, duplicate detection, acknowledgement and retransmission without relying on an existing mesh stack.
- To design and evaluate an adaptive, link-quality-aware routing algorithm that scores each link using RSSI, SNR, packet-loss rate, and battery level, and compares its performance against baseline shortest-hop and flooding routing.
- To implement message confidentiality and node authentication (symmetric encryption + signed node identities) so radio traffic cannot be trivially read or spoofed by a third party.

- To build a companion Android application that lets a user send/receive message, and view connected node, hop count, signal strength, delivery status, and live network topology.
- To quantitatively evaluate the network under realistic conditions multi-hop chains, deliberate node failure, and RF interference using Packet Delivery Ratio, end-to-end delay, throughput, and energy consumption as metrics.

4. Scope

Target users: disaster-response teams, event organizers in low-connectivity venues, hikers/ expedition groups, and rural communities with unreliable cellular coverage.

The final deliverable is a working demonstration of 5-6 physical nodes placed around campus, sending a text message from Phone A to Phone B across multiple hops with no cellular, Wi-Fi, or internet involvement including a live demonstration of automatic route recovery when an intermediate node is powered off.

5. Proposed Methodology

Hardware platform:

Each node combines an ESP32-S3 microcontroller with a Semtech SX1262 LoRa transceiver over SPI, a Li-ion battery with a charge/protection circuit, and BLE for phone connectivity. Optional additions (GPS, OLED status display, solar charging) are treated as stretch goals after the core mesh is proven stable.

Firmware / networking layer:

The ESP32 firmware implements packet framing, node identification, neighbor discovery, store-and-forward relaying, TTL-bounded flooding for discovery, and acknowledgement-based reliable delivery with duplicate suppression.

Adaptive routing:

Each node continuously logs RSSI, SNR, packet-loss rate, and battery percentage for its neighbors and computes a composite link-quality score. The routing layer selects the highest-scoring path instead of the shortest one, and the project experimentally compares this against shortest-hop and pure-flooding routing on the same physical topology.

Mobile application:

An Android app communicates with its nearest node over BLE, providing a chat-style interface plus a live view of network topology, signal strength, hop count, and delivery status.

Security layer:

Symmetric-key message encryption(AES-256-GCM) and per-node signed identifiers prevent casual eavesdropping and node impersonation on the shared radio channel.

Evaluation:

Field tests across direct link, 2-hop, and 4+ hop chains; deliberate node shutdown to test route recovery; and interference/RF-condition tests, each measured against the metrics listed in Objectives.

6. Resources Involved

Hardware:

- ESP32-S3 development boards ×5-6
- SX1262 LoRa/LLCC68/SX1278(alternative) transceiver modules + antennas ×5-6
- Li-ion/LiPo batteries with charging/protection circuitry
- Breadboards, wiring, enclosures (student-budget components, no lab-restricted or specialized equipment)

Software / firmware:

- Arduino / ESP-IDF (C/C++) for node firmware
- Android Studio (Kotlin/Java) for the mobile application
- Python for offline data logging, routing-metric analysis, and result graphing
- Git/GitHub for version control and collaborative development

Reference systems (design inspiration only, not reused as-is):

Meshtastic: open-source LoRa mesh texting firmware, used as a design reference for protocol structure, not cloned wholesale

7. Description of Industrial Support (If any)

8. SDGs (If Applicable)

☐ No Poverty	☐ Zero Hunger
☐ Good Health and Well-Being	☐Quality Education
☐ Gender Equality	☐ Clean water and Sanitation
☐ Affordable and Clean Energy	☐ Decent Work and Economic growth
☒ Industry, Innovations and Infrastructure	☐ Reduced Inequalities
☒ Sustainable Cities and Communities	☐ Responsible Consumption and Production
☐ Climate action	☐ Life Below Water
☐ Life on Land	☐ Peace, Justice and Strong Institutions
☐ Partnerships	

9. Gantt Chart

Year	2026 to 2027								
Months	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Research LoRa/RF/mesh fundamentals	✓								
build and validate a first point-to-point ESP32↔LoRa link.	✓	✓							
BLE bridge to Android; send a message phone-to-phone through two nodes.			✓						
Extend to a 4-node chain; implement discovery, forwarding, TTL, and duplicate detection.				✓					
Reliability layer: ACKs, retransmission, sequence numbers, store-and-forward, route recovery.					✓				
Security layer: encryption, node authentication, message integrity checks.					✓	✓			
Adaptive routing: collect RSSI/SNR/loss/battery data; implement and tune the link-quality metric.						✓			
Comparative evaluation: adaptive vs. shortest-hop vs. flooding routing; optional GPS/topology visualization.							✓		
Field testing: direct link, multi-hop, node-failure, interference, and battery-life tests across campus.								✓	
Results, graphs, documentation, thesis write-up, poster, and final demonstration									✓

10. Details of Project Team

i. Students

No.	Name	Seat No.	Signature (s)
1	Shayan Malik	CS-23053	
2	Syyeda Fatima Shoaib	CS-23075	
3	Umaiza Asif	CS-23080	
4	Hamna Rizvi	CS-23082	

ii. Supervisors / Advisors

	Name	Designation & Department	Address & Contact	Signature(s)
Supervisor	Dr. Sumayya Zafar	Assistant Professor	0300-2640813 CISD	
Co-Supervisor (If any)	Sheikh Abdul Basit	Team Lead NCCS Cybersecurity Instructor, SSU	0303-0378946 NCCS	
Industrial Advisor (If any)				

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Project Serial No.: _____ Dated: _____	Signature Convener Steering Committee	Signature FYP Coordinator

☐ Proposal Approved	☐ Not Approved	☐ Returned for Clarification / Modification
Comments: (if any)		

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(Signature of Chairperson)

Date: _____